1. **Answer:** $\boldsymbol{9\pi}$
2. **Answer:** $\boldsymbol{10\pi}$
3. **Answer:** $\boldsymbol{13\sqrt{3} - \frac{4\pi}{3}}$
4. **Answer:** $\boldsymbol{\sqrt{37}}$
5. **Answer:** $\boldsymbol{\frac{66}{25}}$
6. **Answer:** $\boldsymbol{\frac{448}{15}}$
7. **Answer:** $\boldsymbol{\frac{48}{5}}$
8. **Answer:** $\boldsymbol{\frac{81}{20}}$
9. **Answer:** $\boldsymbol{128}$
10. **Answer:** $\boldsymbol{\frac{343\sqrt{2}}{16}}$